



NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 -Electricity Markets and Energy System Planning, Specialisation Course

Hydropower Scheduling Hierarchy and Water Values

Hossein Farahmand
Professor
Autumn 2026



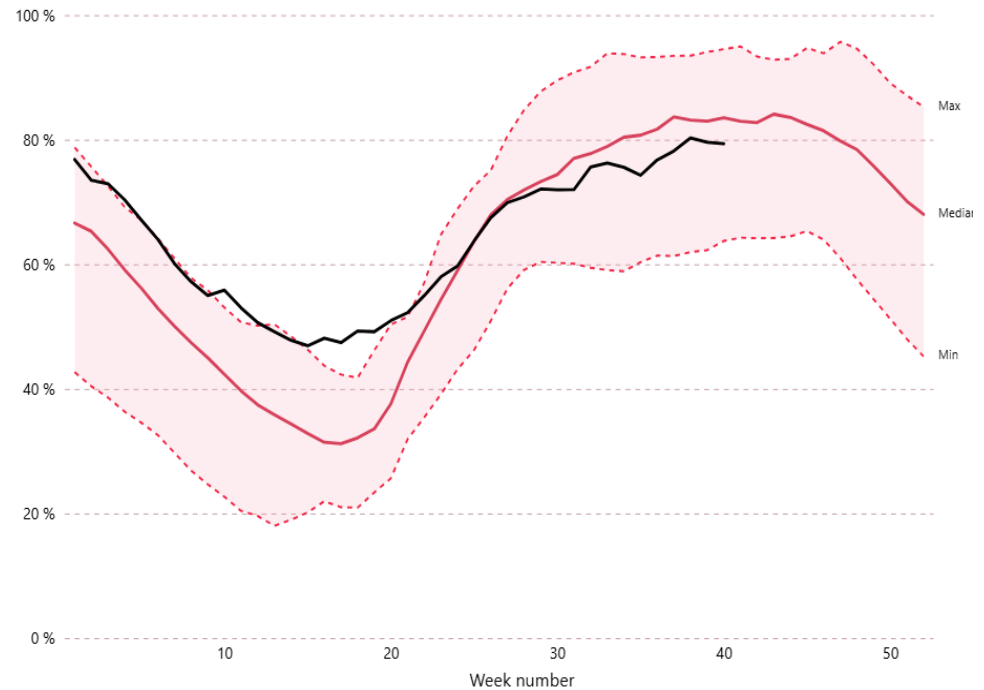
Overview

Hydropower Scheduling Hierarchy

Water Value Calculation



Percentage filling (%) Norway, 2025

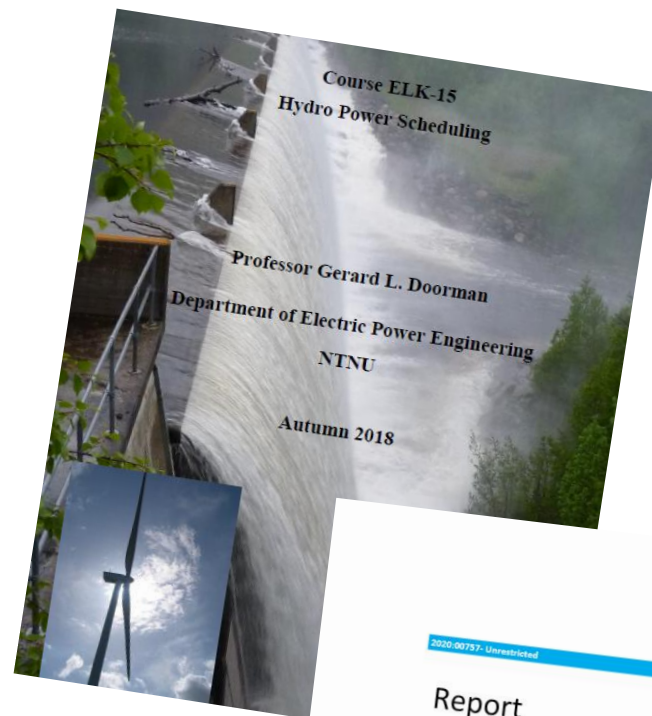


The figures represent 97.5 percent of the total reservoir capacity. The total reservoir capacity is 86.9 TWh. Pr. 01.01.2017: 86920 GWh. Min, max and median values for the period 2000 - 2019.

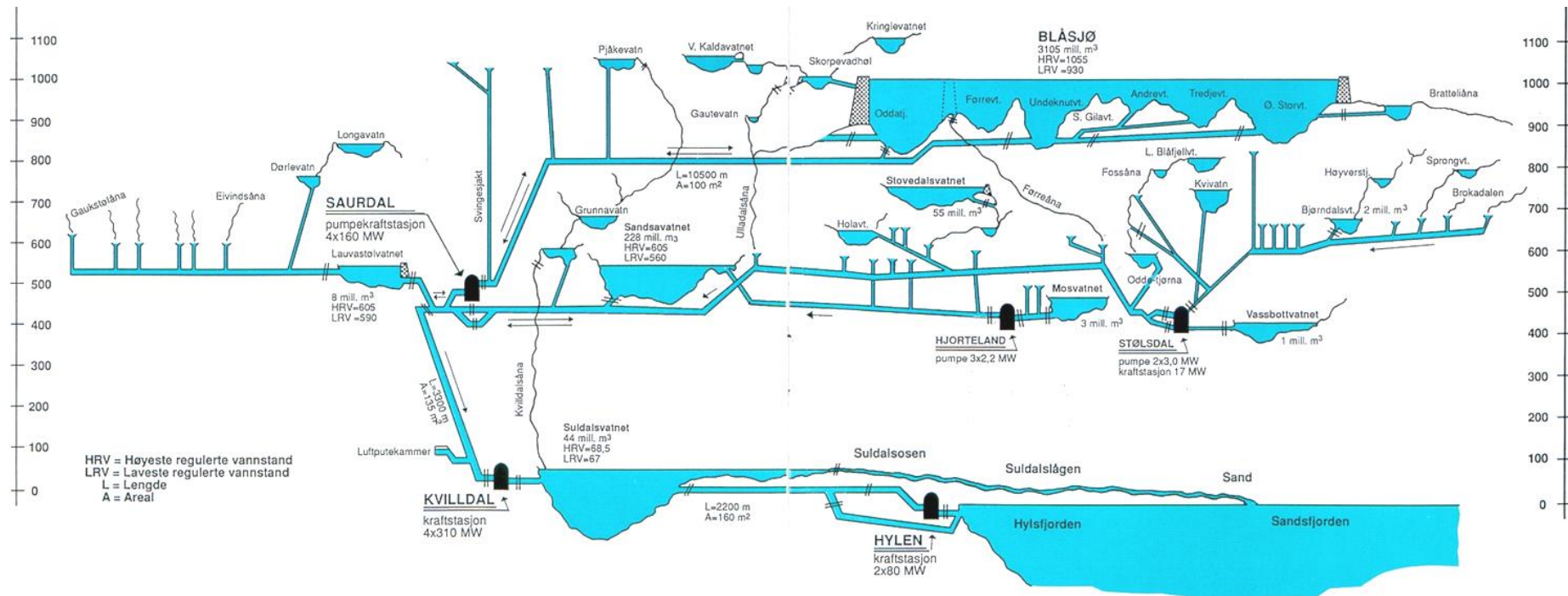
Source: [The Norwegian Water Resources and Energy Directorate \(NVE\)](https://www.nve.no/)

References

- Compendium in hydro power scheduling
- Scheduling Toolchains in Hydro-Dominated Systems
- Found in the “Learning materials” folder on the blackboard.



Storage and waterways-Real life example



--> Complex Storage Scheme:

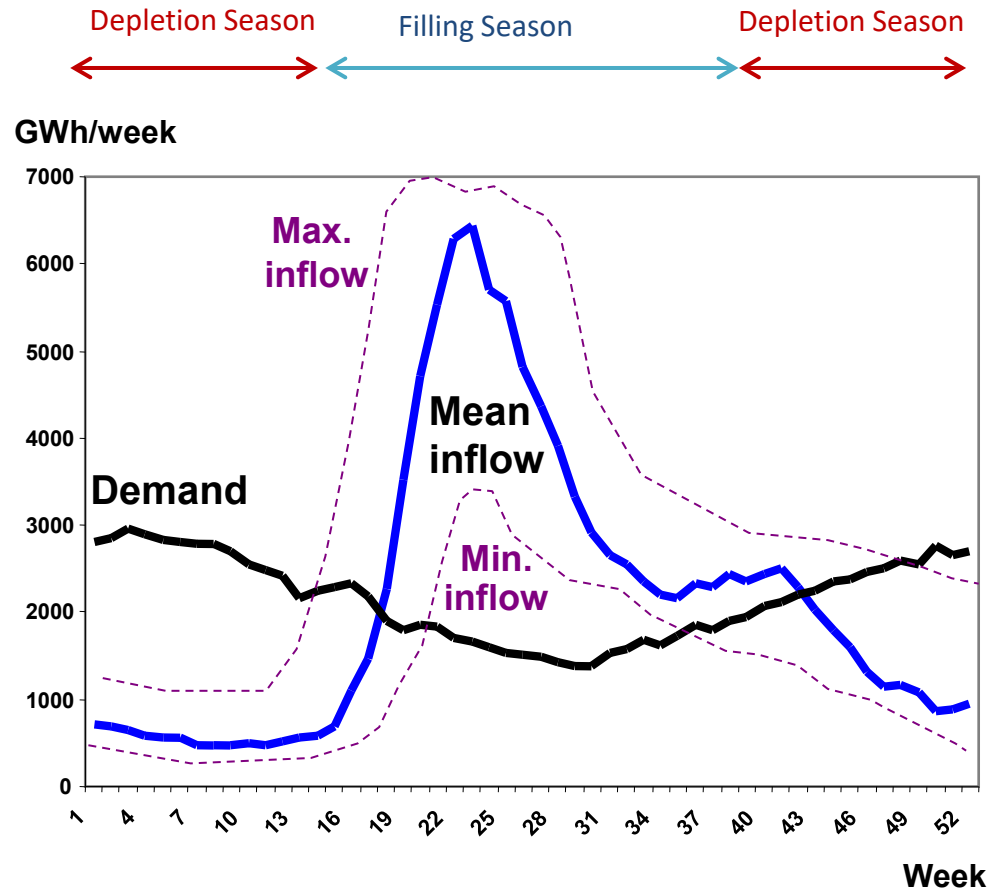
- 1 Major reservoir, contains water for multi-year production (in case of dry year(s))
- 24 smaller reservoirs that are channeled into the system
- 3 Major Power plants (all underground), and 2 pumping stations

Scheduling Task

Different time horizons

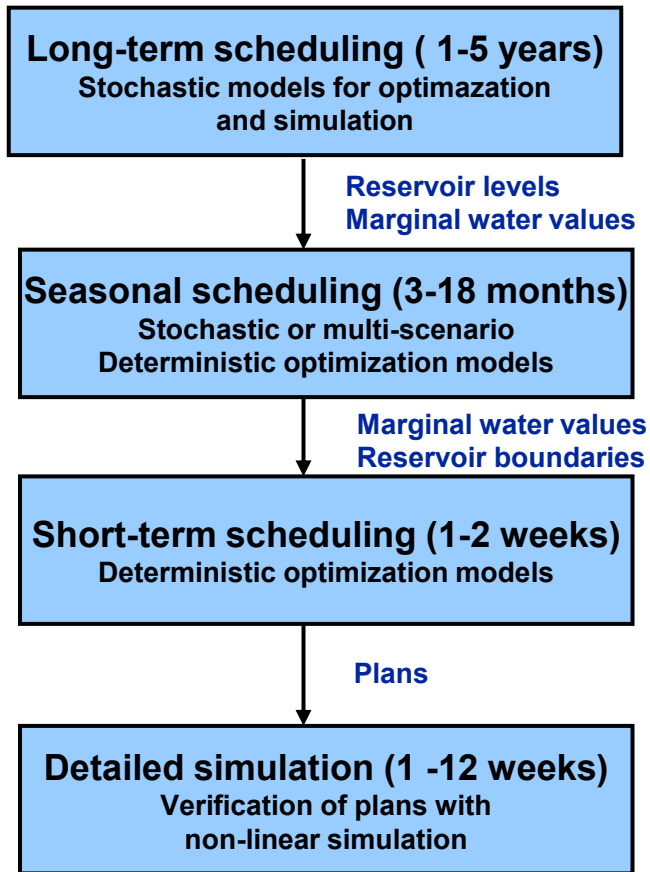
Different geographical scope

Different degree of details

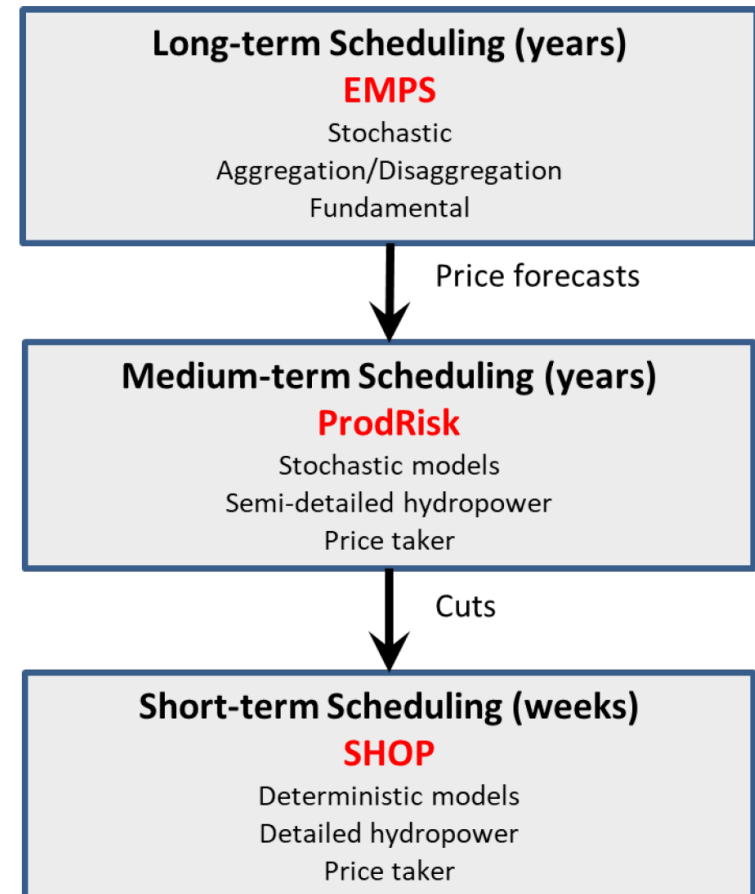


Scheduling Hierarchy

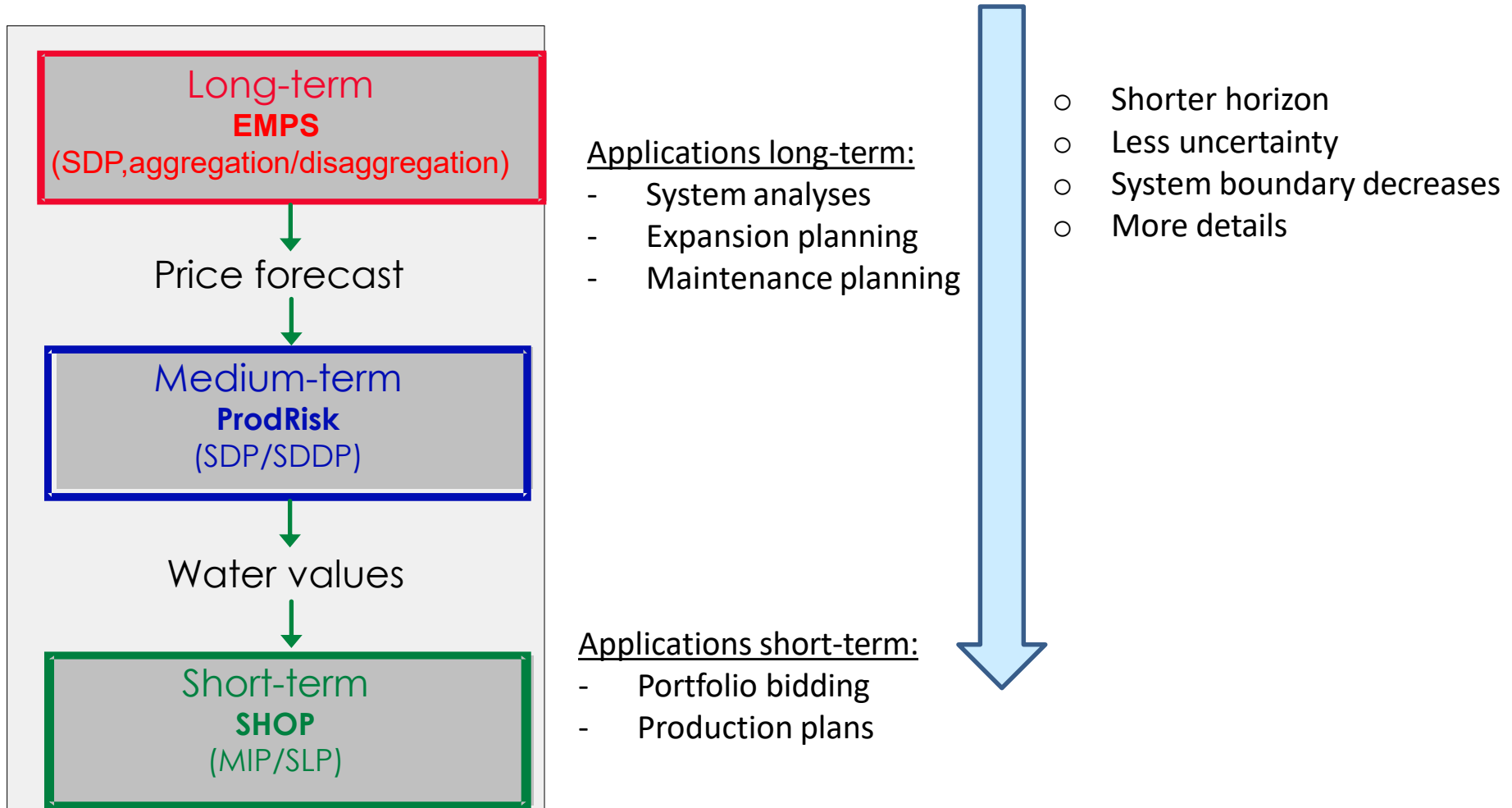
Compendium



A. Helseth and A. C. Geber de Melo, "Scheduling Toolchains in HydroDominated Systems: Evolution, Current Status and Future Challenges for Norway and Brazil," SINTEF Energy Research, Tech. Rep., 2020, <https://sintef.brage.unit.no/sintef-xmlui/handle/11250/2672581?locale-attribute=en>.



Scheduling Hierarchy



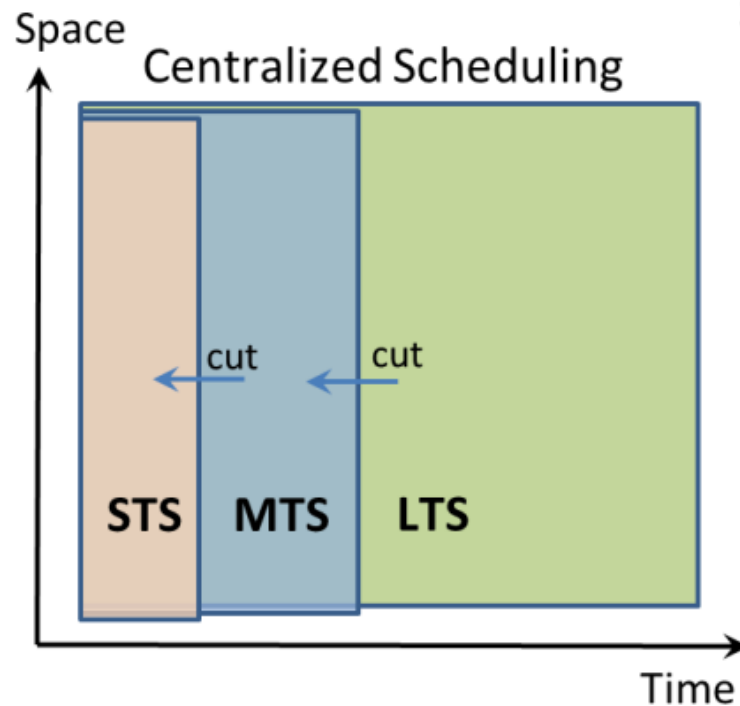
A. Helseth and A. C. Geber de Melo, "Scheduling Toolchains in HydroDominated Systems: Evolution, Current Status and Future Challenges for Norway and Brazil," SINTEF Energy Research, Tech. Rep., 2020, <https://sintef.brage.unit.no/sintef-xmlui/handle/11250/2672581?locale-attribute=en>.

Two (equivalent) objectives

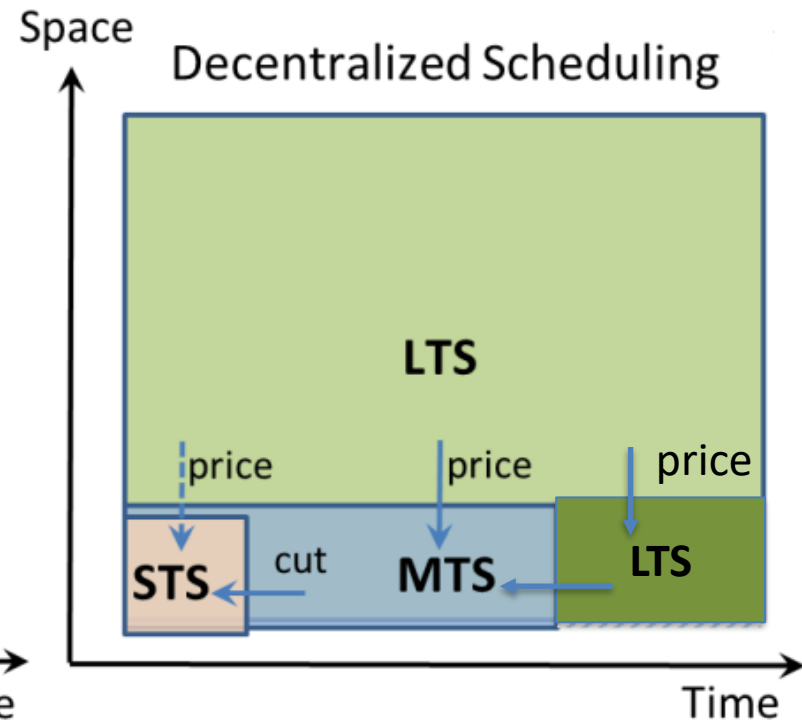
- Cost minimization
 - Relevant in centralized markets (Brazil, Iceland) and Norway before deregulation (1991)
 - Objective: minimize expected operational costs over time
- Profit maximization
 - Relevant in decentralised markets
 - Market simulation: Prices
 - Profit maximization per company

Geographical scopes

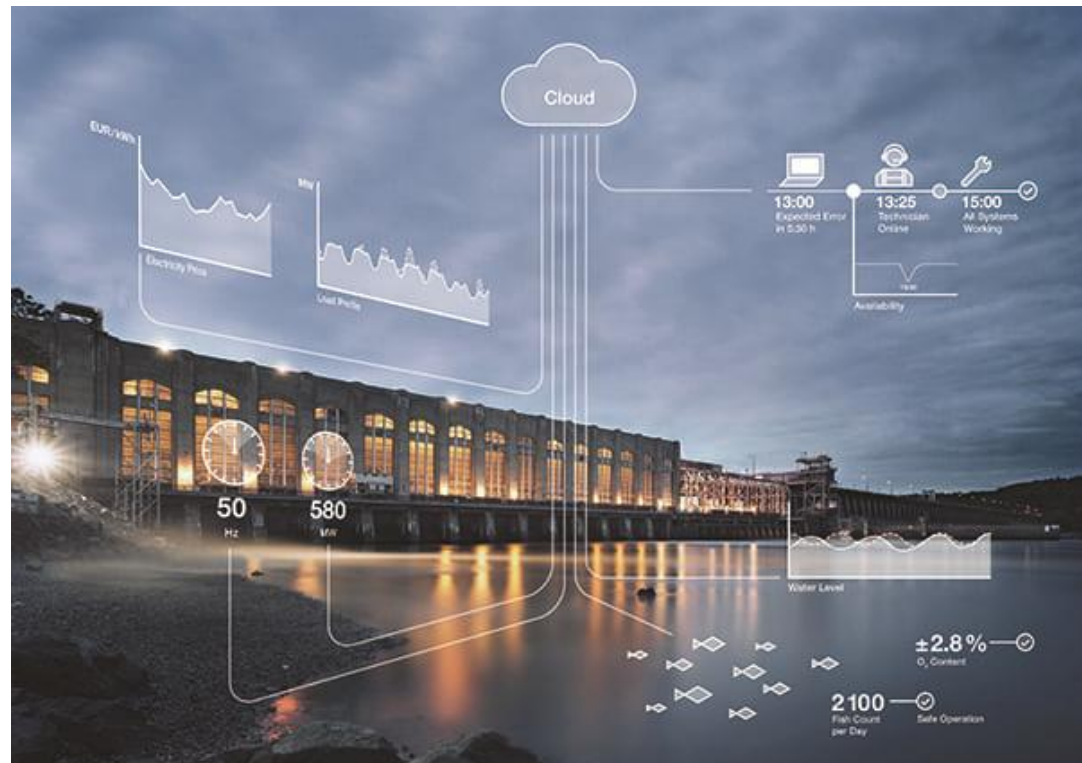
Example: Brazil



Example: Norway



Water Values

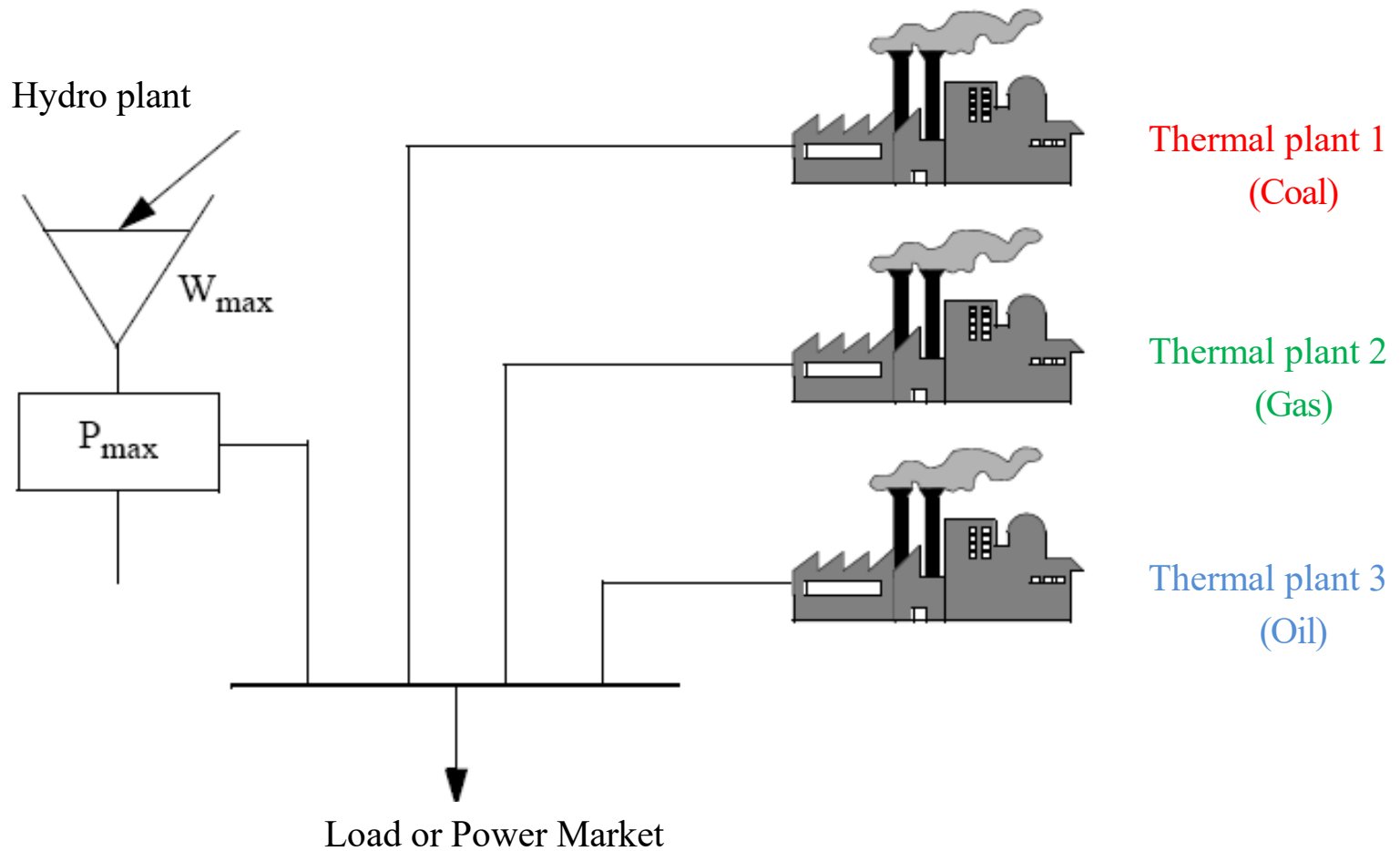


Water value – Definition

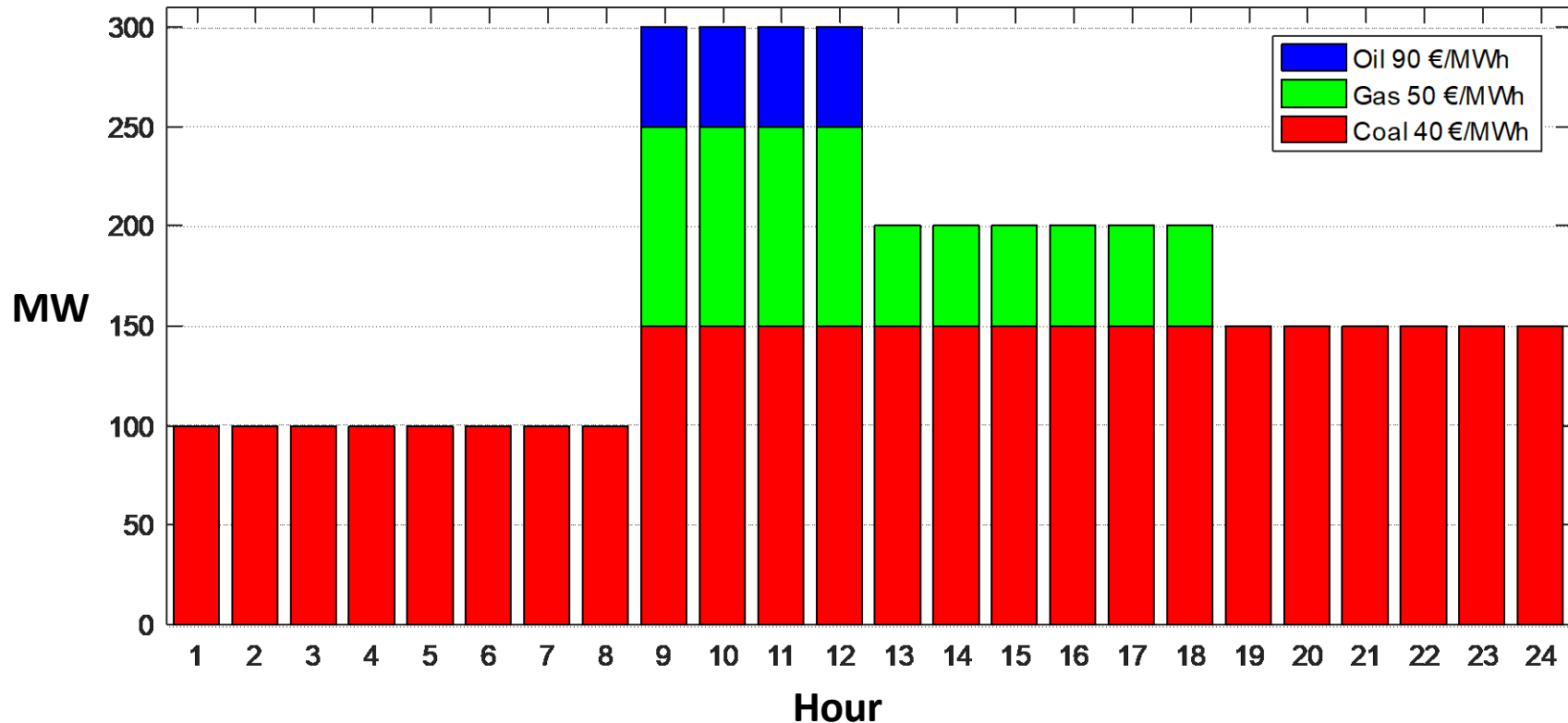
The ***water value*** is the ***expected marginal value*** of having an additional unit of water in the reservoir

We will use Stochastic Dynamic Programming to find the water value

Deterministic Example hydro-thermal system

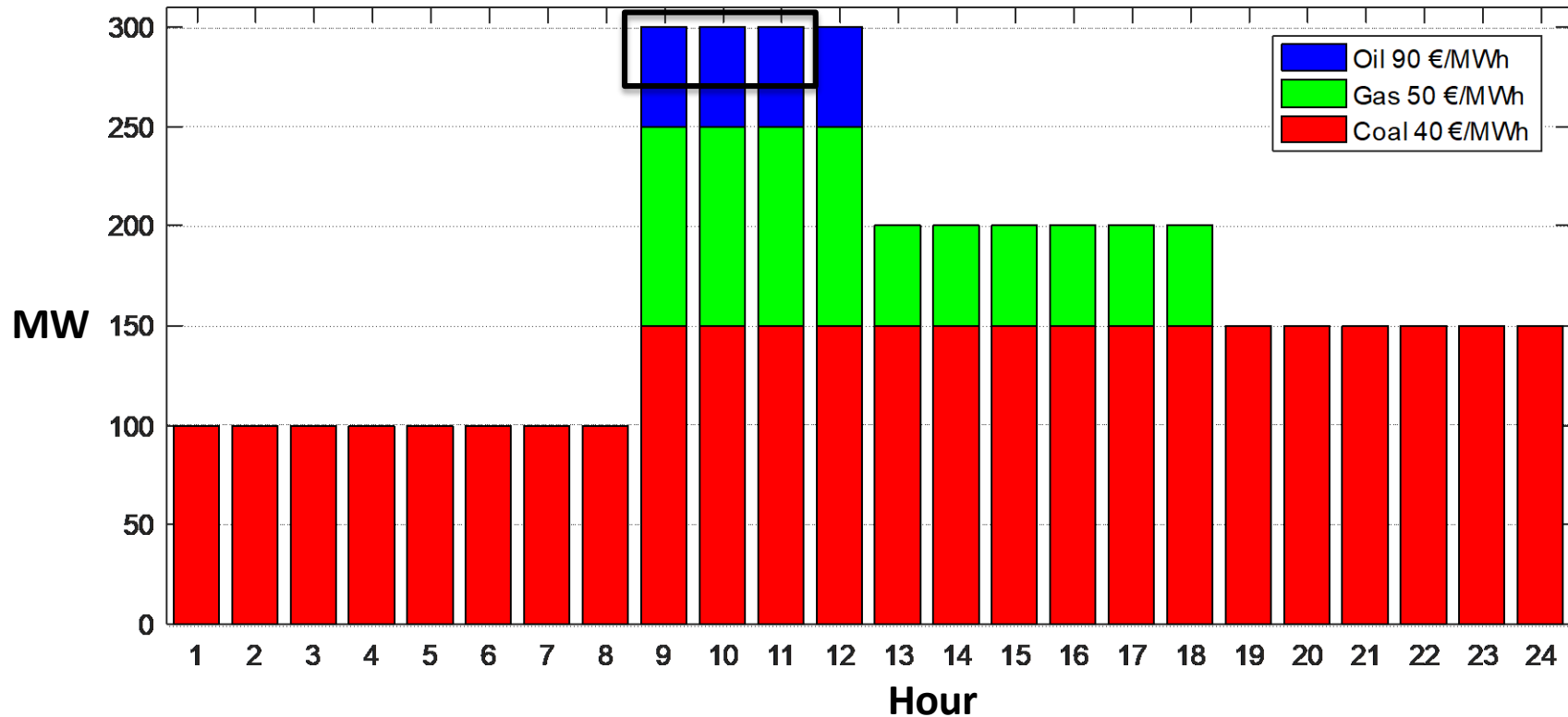


Water Value Calculation-Example



- Hydro: 25 MW, 75 MWh
 - How to use it optimally?
 - Marginal value of having 1 more MWh?

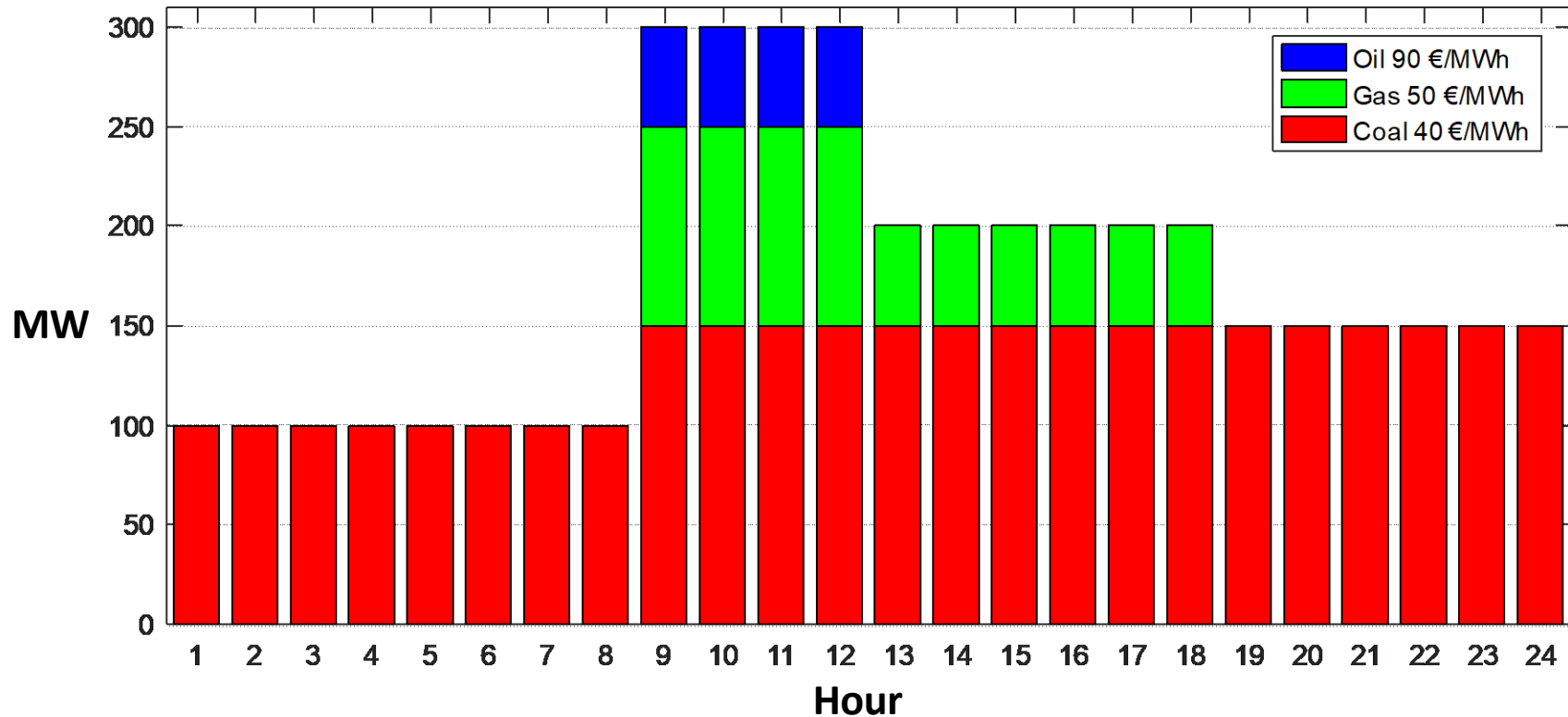
Water Value Calculation-Example (1)



- Hydro: 25 MW, 75 MWh
 - How to use it optimally?
 - Marginal value of having 1 more MWh?

Hydro replaces Oil -> Water value is 90 €/MWh

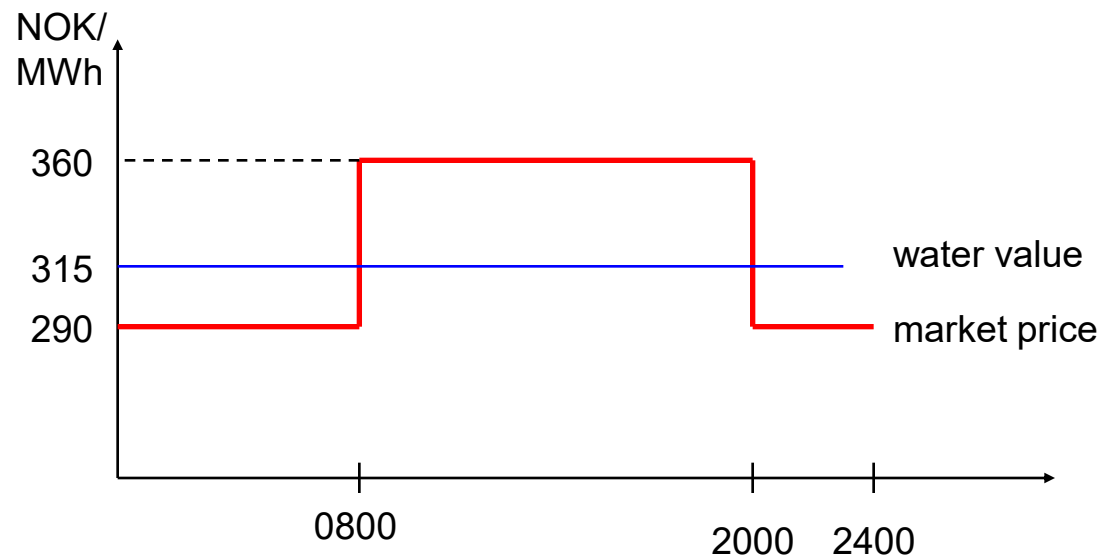
Water Value Calculation-Example (2)



- Find the water value of the following hydro power specifications:
 - A) 25 MW power capacity and 100 MWh stored energy **50**
 - B) 25 MW power capacity and 300 MWh stored energy **40**
 - C) 100 MW power capacity and 150 MWh stored energy **90**

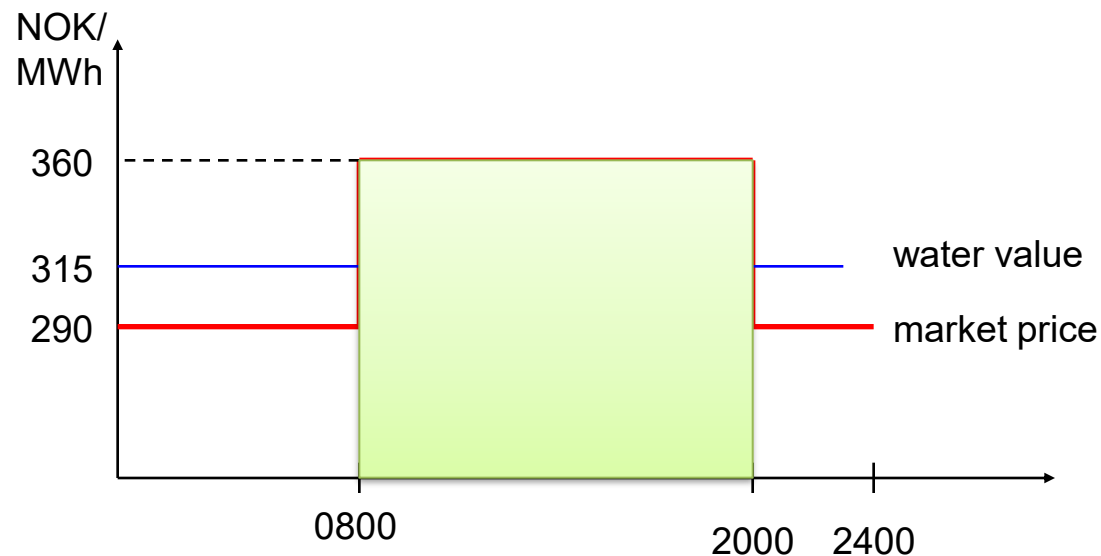
Optimal Hydropower Dispatch– example

- Water value: 315 NOK/MWh
- Plant capacity 100 MW
- Market price:

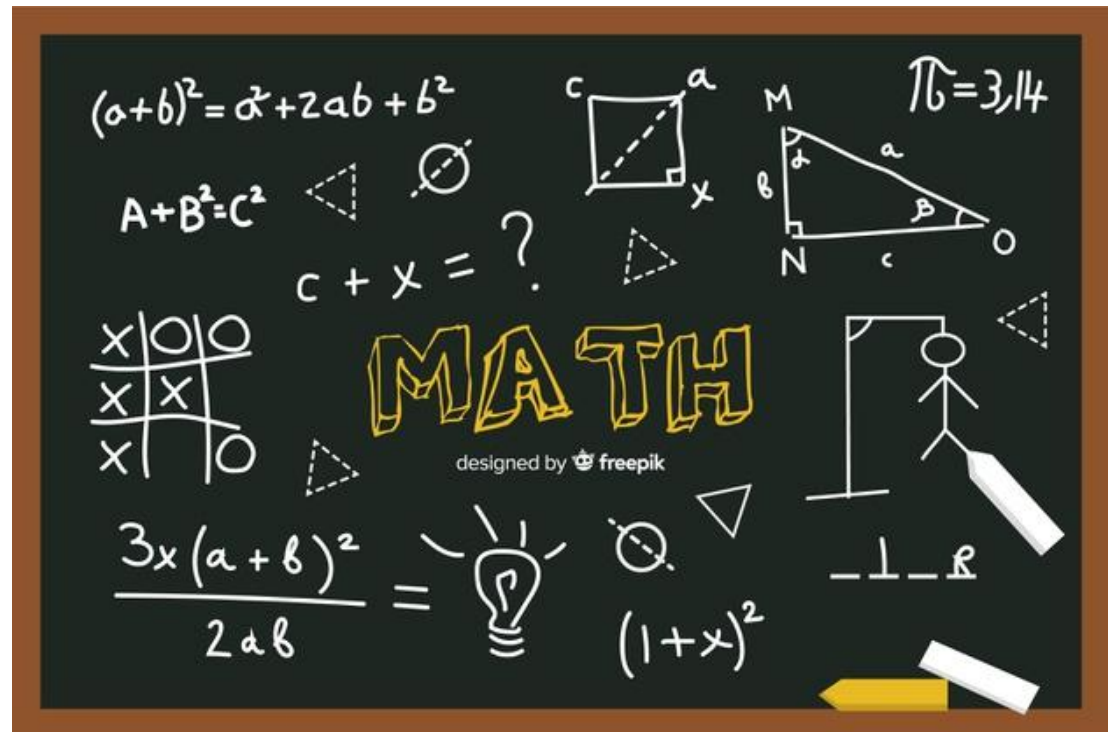


Optimal Hydropower Dispatch– example

- Water value: 315 NOK/MWh
- Plant capacity 100 MW
- Market price:



Water Value Calculation



Mathematical background-recap from previous lecture

Consider the following two-stage deterministic problem:

$$\min_{x_1, x_2} \underbrace{c_1 x_1}_{\text{First-stage cost}} + \underbrace{c_2 x_2}_{\text{Second-stage cost}}$$

subject to

$$A_1 x_1 \geq b_1 \quad \leftarrow \text{First-stage constraint}$$

$$\underbrace{E_1 x_1 + A_2 x_2}_{\text{Linking constraint}} \geq b_2$$

Mathematical background

Solve the first-stage and the second-stage problems separately

- **Step 1)** Discretize x_1 into a set of trial segment values $\{\hat{x}_{1i}, i = 1, \dots, n\}$
- **Step 2)** Solve the second-stage problem for each trial value
- **Step 3)** Construct cost function $\alpha(x_1)$.
Intermediate values of $\alpha(x_1)$ are obtained by interpolation from the neighboring discretized values.
- **Step 4)** Solve the first-stage problem using the future cost function constructed

First-stage problem:

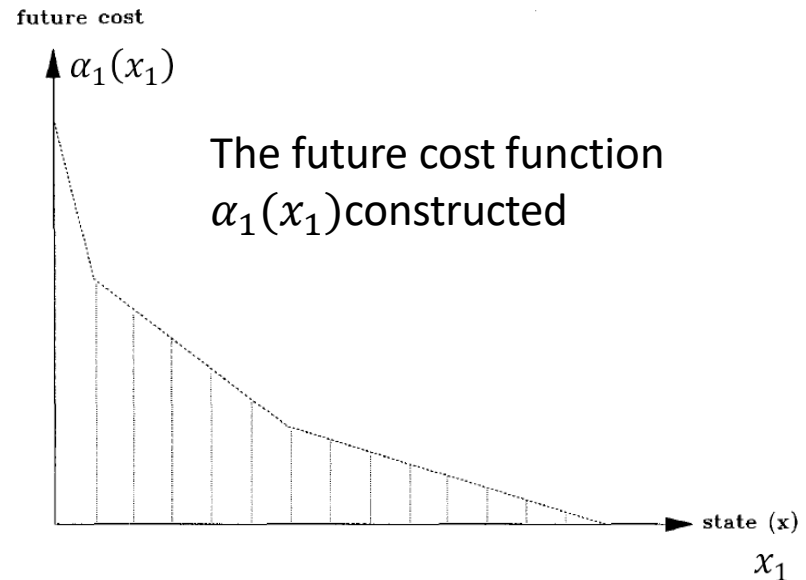
$$\begin{aligned} & \min_{x_1} c_1 x_1 + \alpha_1(x_1) \\ & \text{subject to} \\ & A_1 x_1 \geq b_1 \end{aligned}$$

Second-stage problem:

$$\begin{aligned} & \alpha_1(x_1) = \min_{x_2} c_2 x_2 \\ & \text{subject to} \\ & A_2 x_2 \geq b_2 - E_1 x_1 \end{aligned}$$

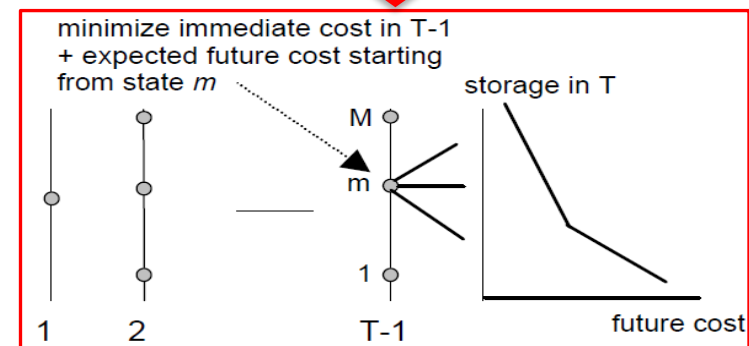
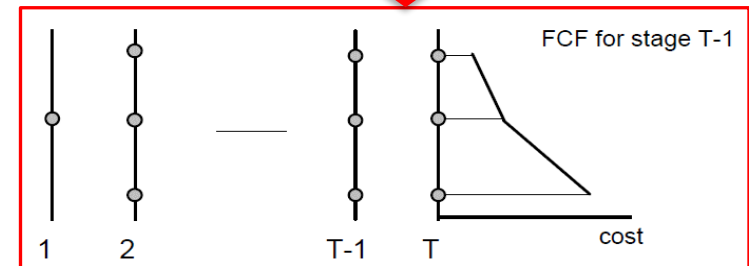
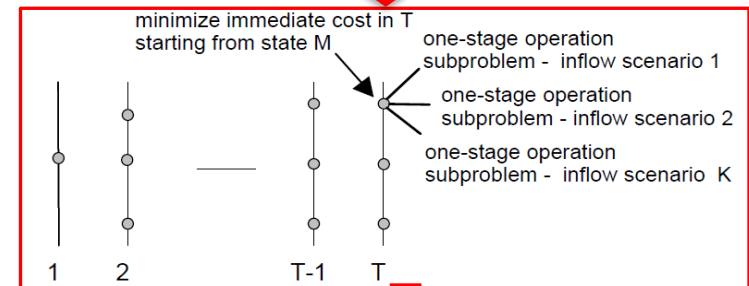
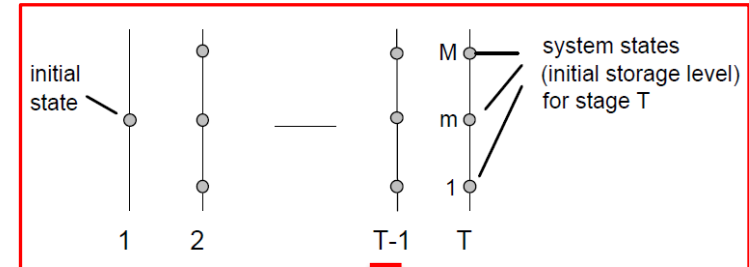
Dynamic programming (DP)

- **Step 1)** Discretize x_1 into a set of trial segment values $\{\hat{x}_{1i}, i = 1, \dots, n\}$
- **Step 2)** Solve the second-stage problem for each trial value
- **Step 3)** Construct cost function $\alpha(x_1)$. Intermediate values of $\alpha(x_1)$ are obtained by interpolation from the neighboring discretized values.
- **Step 4)** Solve the first-stage problem using the future cost function constructed



SDP Recursion

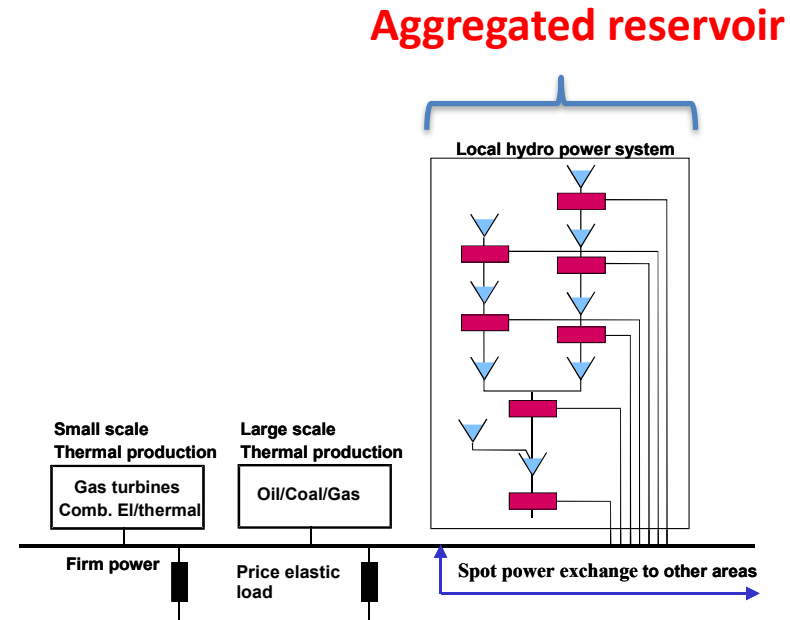
- **Step a)** Definition of system states for each stage t , discretize the system states (e.g., reservoir storage levels) into several segments to represent different operating conditions. The initial state is assumed to be known.
- **Step b)** Optimal Strategy Calculation - Last Stage
- **Step c)** Calculation of the FCF for Stage T-1
- **Step d)** Backward recursion for previous stages: Calculation of Operation Costs for Stage T-1 and FCF for Stage T-2
- **Step e)** Continue the backward recursion until the values of the Future Cost Function (FCF) in the first and last stages (or consecutive iterations) are approximately equal.
- **Step f)** Forward simulation and update of terminal values



Water value method

Stochastic Dynamic Programming (SDP)

- **Stage:** time – one week
- **State:** reservoir volume (discretized, 51 levels)
- **Return function:** operation dependent cost
- **Decision:** hydro production (energy “withdrawal” from reservoir)



Nomenclature

Sets

t – timesteps [weeks]

Parameters

C^g - Cost of grid import

C^r - Cost of rationing

D_t - Consumption

Stochastic variables

l_t - inflow [MWh/h]

Decision variables

g_t - grid import [MWh/h]

r_t - rationing [MWh/h]

p_t - hydro production [MWh/h]

x_t - reservoir level at start of period [Mwh]

s_t - spill [Mwh]

Functions

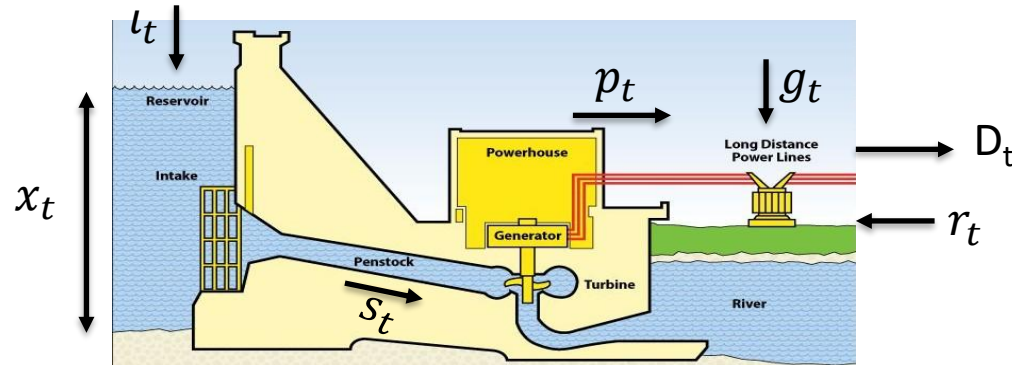
FC – future cost

Dual multipliers

λ_t - marginal cost of operation

μ_t – marginal value of stored water

Problem formulation for deterministic single stage



$$\min_{g, r, x, p, s \geq 0} J_t = C^g g_t + C^r r_t + F C_{t+1}(x_{t+1}) \quad \text{Objective function}$$

Subject to:

$$\begin{aligned} p_t + g_t + r_t &= D_t & (\lambda_t) & \text{Power balance constraint} \\ X_t - p_t - s_t + l_t &= x_{t+1} & (\mu_t) & \text{Reservoir balance constraint} \\ g_t &\leq G^{\max} \\ p_t &\leq P^{\max} \\ X^{\min} &\leq x_t \leq X^{\max} \end{aligned}$$

SDP for Hydropower

Goal:

Optimize multi-stage reservoir operation under uncertain inflows (ι_t), and demands (D_t).

Stage t : Decision problem

$$\min_{g,r,x,p,s \geq 0} J_t = C^g g_t + C^r r_t + FC_{t+1}(x_{t+1})$$

Subject to:

$$\begin{aligned} p_t + g_t + r_t &= D_t & (\lambda_t) \\ X_t - p_t - s_t + \iota_t &= x_{t+1} & (\mu_t) \\ g_t &\leq G^{\max}, \quad p_t \leq P^{\max}, \quad X^{\min} \leq x_t \leq X^{\max} \end{aligned}$$

Algorithm (pseudocode):

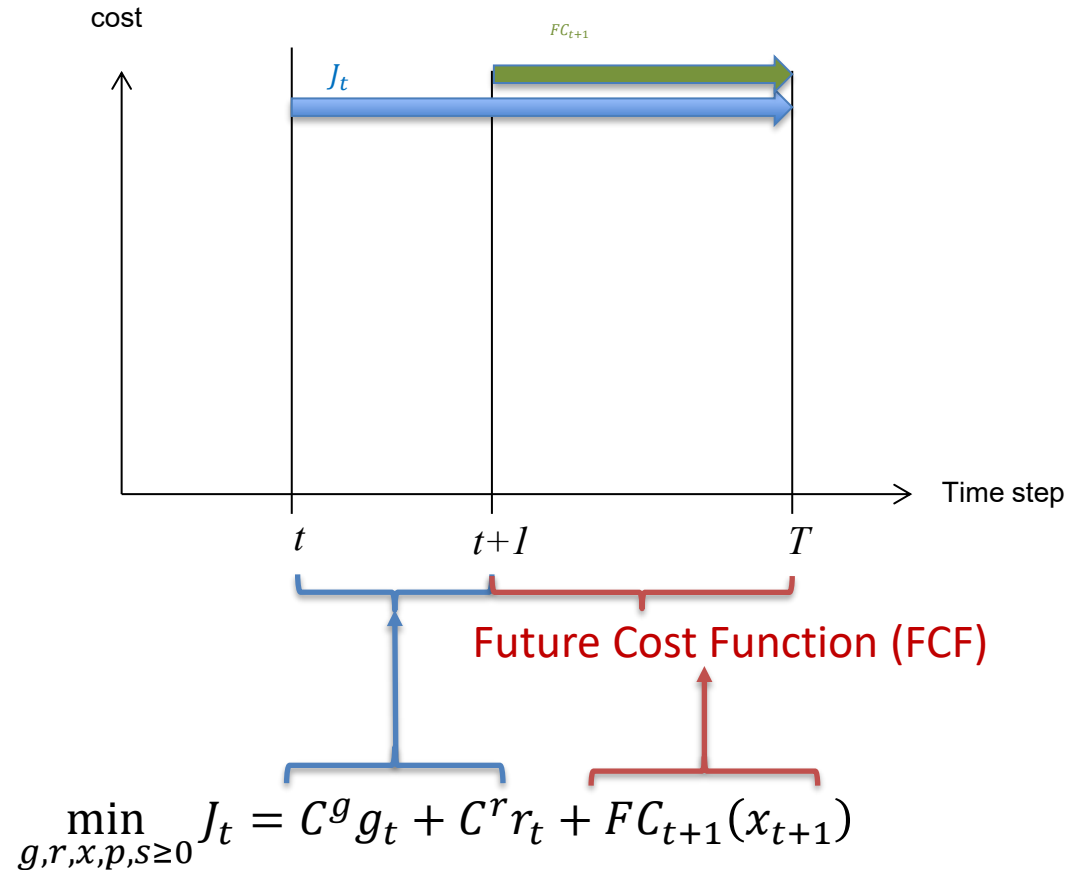
1. **Initialization**: Set terminal value $V_{T+1}(x_{T+1}) = FC_{T+1}(x_{T+1})$
2. **Backward recursion**: For $t = T \rightarrow 1$, for all reservoir levels x_t compute:

$$V_t(x_t) = \min_{g,r,x,p,s \geq 0} \left[C^g g_t + C^r r_t + \mathbb{E}_{\iota_t} [V_{t+1}(x_{t+1})] \right]$$

3. **Forward simulation**: Given inflow realizations ι_t , apply optimal policy to evaluate energy production, revenue, and final storage.

Outputs: Optimal releases p_t^*, r_t^*, s_t^* , value functions $V_t(x_t)$, and marginal water values $\frac{\partial V_t}{\partial x_t}$.

Objective function



$$\min_{g,r,x,p,s \geq 0} J_t = C^g g_t + C^r r_t + J_{t+1} \quad \text{Bellman formulation}$$

Formulate the optimisation problem as a Lagrangian problem

$$\min_{g_t, r_t, p_t, x_{t+1}, \lambda_t, \mu_t} L = \begin{cases} C^g g_t + C^r r_t + F C_{t+1}(x_{t+1}) \\ + \lambda_t (p_t + g_t + r_t - D_t) \\ + \mu_t (X_t - p_t - s_t + l_t - x_{t+1}) \end{cases}$$

- The Karush-Kuhn-Tucker condition says that either the dual multiplier must be 0, or the restriction must be binding.
- The power balance restrictions and reservoir restrictions are equations and per definition binding.

Water value - calculation

- Water value = the marginal value of having one extra unit stored in the reservoir
- Equivalent to μ_t the dual multiplier of the reservoir constraint.
- Note that μ_t refers to the reservoir filling of time t .

Hydro power system optimality

$$\frac{\partial L}{\partial p_t} = \lambda_t - \mu_t = 0$$

$$\frac{\partial L}{\partial x_{t+1}} = \frac{\partial FC_{t+1}}{\partial x_{t+1}} - \mu_t = 0$$

$$\frac{\partial FC_{t+1}}{\partial x_{t+1}} = \lambda_t$$

- The water value has to be equal to the marginal cost of operation
- In a perfect competitive market equivalent to
 - Price = water value at end of period
- Trade off: Use water now for lower cost, or save water for lower cost later
- Note: Assuming no other binding constraints.

Envelope Theorem (sketch of proof)

The Envelope Theorem tells you that the partial derivative of the optimal value of the objective function with respect to a parameter is equal to the partial derivative of the objective function with respect to that parameter, evaluated at the optimal values of the decision variables.

$$\begin{array}{ll}
 F(x) = \min_x \{f(x, u)\} & \\
 \text{s.t. } g(x, u) = 0 & (\lambda)
 \end{array}
 \quad \xRightarrow{\text{Postulate}} \quad
 \frac{\partial F}{\partial u} = \frac{\partial f^*}{\partial u} + \lambda \frac{\partial g^*}{\partial u} \quad \text{and } x = x(u)$$

Sketch of proof:

$$L = f(x, u) + \lambda g(x, u) \quad \text{f.o.c.:} \quad \frac{\partial L}{\partial x} = \frac{\partial f^*}{\partial x} + \lambda \frac{\partial g^*}{\partial x} = 0$$

$$\begin{aligned}
 \frac{\partial F}{\partial u} &= \frac{\partial L}{\partial u} = \frac{\partial f}{\partial u} + \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial \lambda}{\partial u} g + \lambda \left(\frac{\partial g}{\partial u} + \frac{\partial g}{\partial x} \frac{\partial x}{\partial u} \right) = \\
 &\quad \frac{\partial f}{\partial u} + \frac{\partial \lambda}{\partial u} g + \frac{\partial x}{\partial u} \left(\overbrace{\frac{\partial f}{\partial x} + \lambda \frac{\partial g}{\partial x}}^{=0} \right) + \lambda \frac{\partial g}{\partial u} = \underline{\underline{\frac{\partial f}{\partial u} + \lambda \frac{\partial g}{\partial u}}} \\
 &\quad \text{Q.E.D.}
 \end{aligned}$$

(indirect derivatives cancel out)

Envelope Theorem

Using the sketch of proof of the envelope theorem, we get

$$\min_{g,r,x,p,s \geq 0} J_t = C^g g_t + C^r r_t + FC_{t+1}(x_{t+1})$$

s.t:

$$\begin{aligned} p_t + g_t + r_t &= D_t & (\lambda_t) \\ X_t - p_t - s_t + \iota_t &= x_{t+1} & (\mu_t) \text{ (which is the only place } X_t \text{ occurs)} \end{aligned}$$

Derivation and the envelope theorem then gives:

$$\frac{\partial J_t}{\partial x_t} = 0 + \mu_t \Rightarrow \frac{\partial J_t}{\partial x_t} = \mu_t$$

$$\frac{\partial J_t}{\partial x_t} = \mu_t \text{ and for } t+1: \frac{\partial J_{t+1}^*}{\partial x_{t+1}} = \mu_{t+1} \Rightarrow \frac{\partial FC_{t+1}^*}{\partial x_{t+1}} = \mu_{t+1}$$

System optimization – optimality for hydropower

Optimality condition

$$\frac{\partial FC_{t+1}}{\partial x_{t+1}} = \lambda_t$$

Water value next period = power price

Envelope theorem applied on t+1 optimization problem

$$\frac{\partial FC_{t+1}^*}{\partial x_{t+1}} = \mu_{t+1}$$

(indirect derivatives cancel out)

Optimality condition (next period)

$$\frac{\partial L}{\partial p_{t+1}} = \lambda_{t+1} - \mu_{t+1} = 0 \quad \Rightarrow$$

$$\lambda_{t+1} = \lambda_t$$

Same price next period

without other binding constraints!

Influence of reservoir restrictions I

$$\min_{g,r,x,p,s \geq 0} C^g g_t + C^r r_t + F C_{t+1}(x_{t+1})$$

Subject to

$$\begin{aligned} p_t + g_t + r_t &= d_t \\ X_t - p_t - s_t + l_t &= x_{t+1} \\ g_t &\leq G^{max} \\ p_t &\leq P^{max} \\ X^{min} &\leq x_t \leq X^{max} \end{aligned}$$

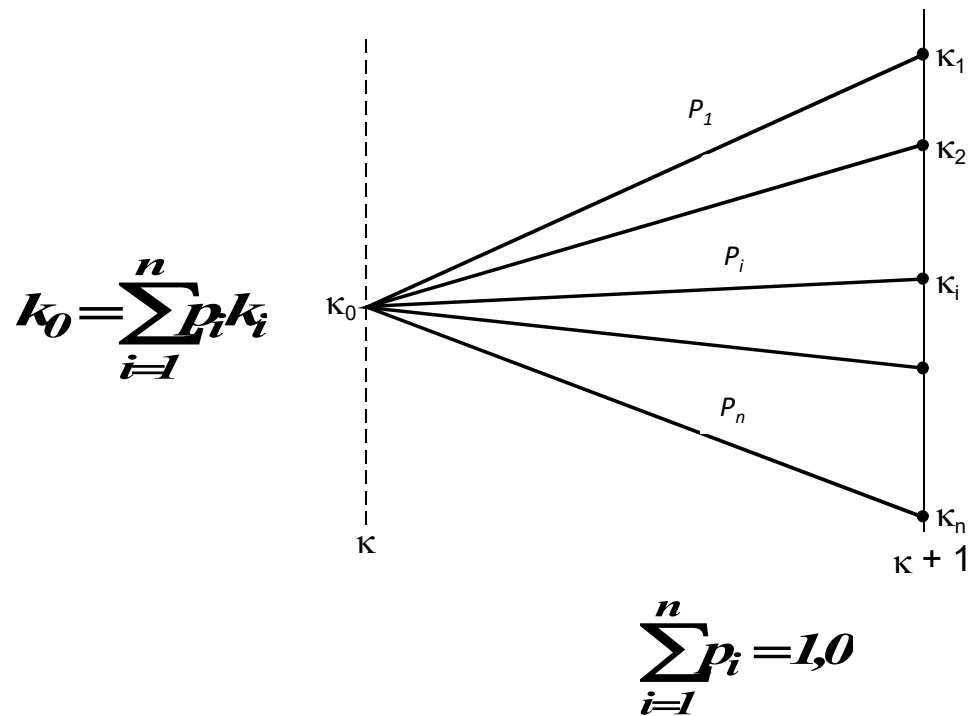
If these restrictions are binding, the lagrangian will have another non-zero term

Note that the reservoir constraint and production limitation drive a wedge between the prices in weeks t and $t+1$

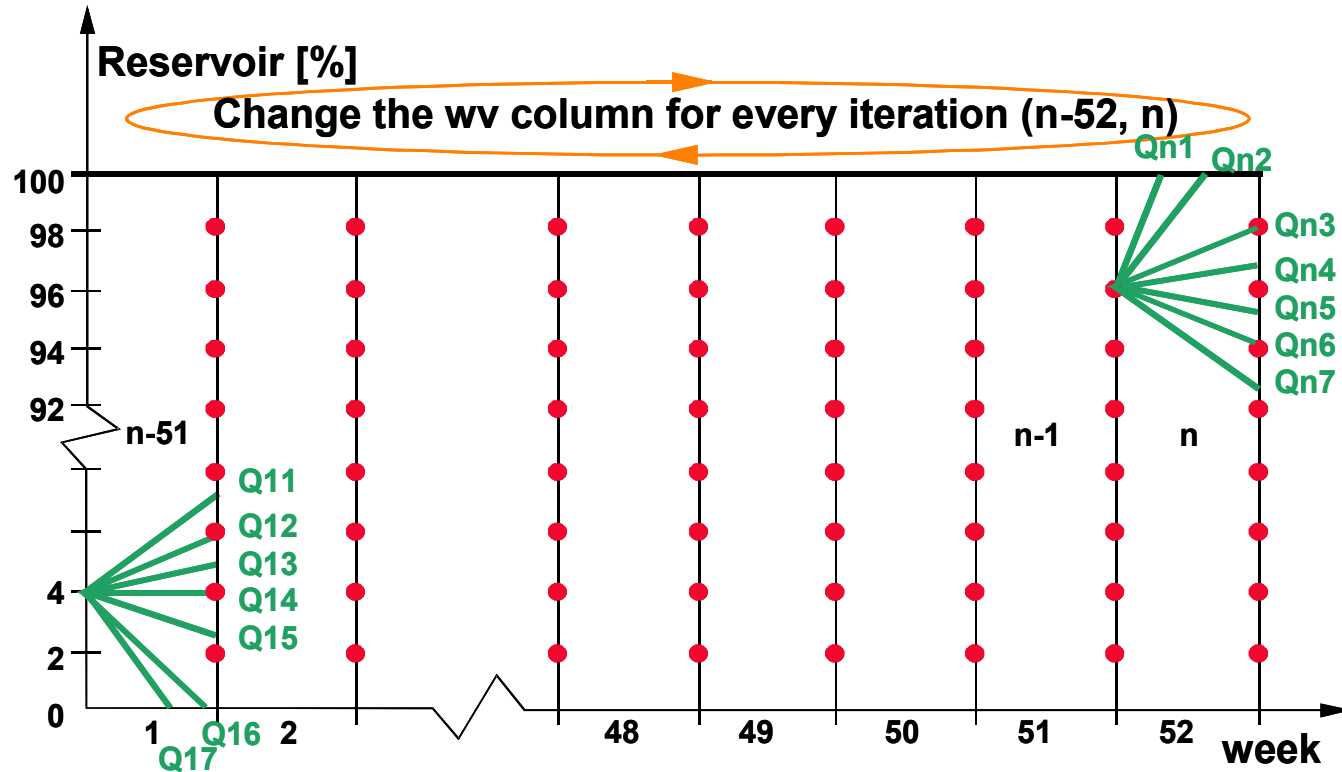
Final values and horizon

- The water value calculation is a form for backward DP where the derivative of the total cost is used as criterion
- When calculating week t , the water values in week $(t+1)$ are assumed known
- The final values must be assumed / estimated
- Approach
 - Estimate final week (end-of-year) values
 - Calculate the operating cost for the final week of the year
 - Propagate backward: calculate values successively from the last week to the first year
 - Set end-of-year values equal to start-of-year values
 - Terminate when convergence otherwise, calculate earlier years
 - Forward simulation and update of each week terminal values and continue from the second bullet

Stochastic inflow

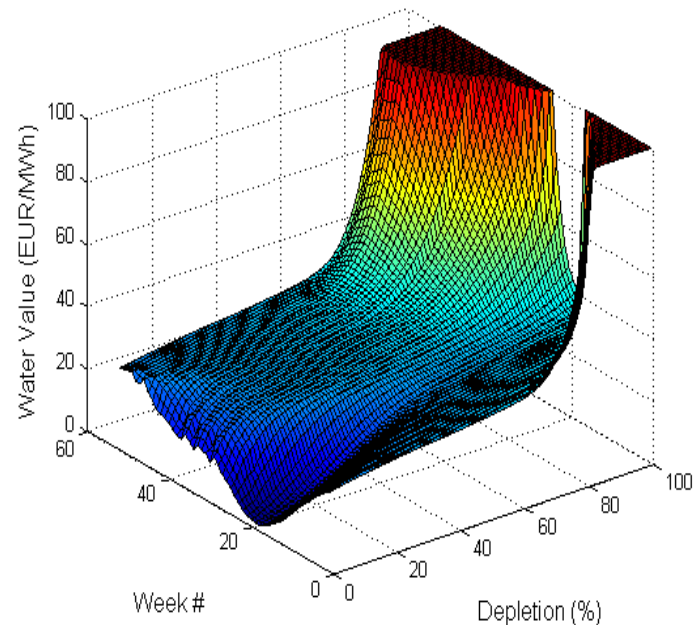
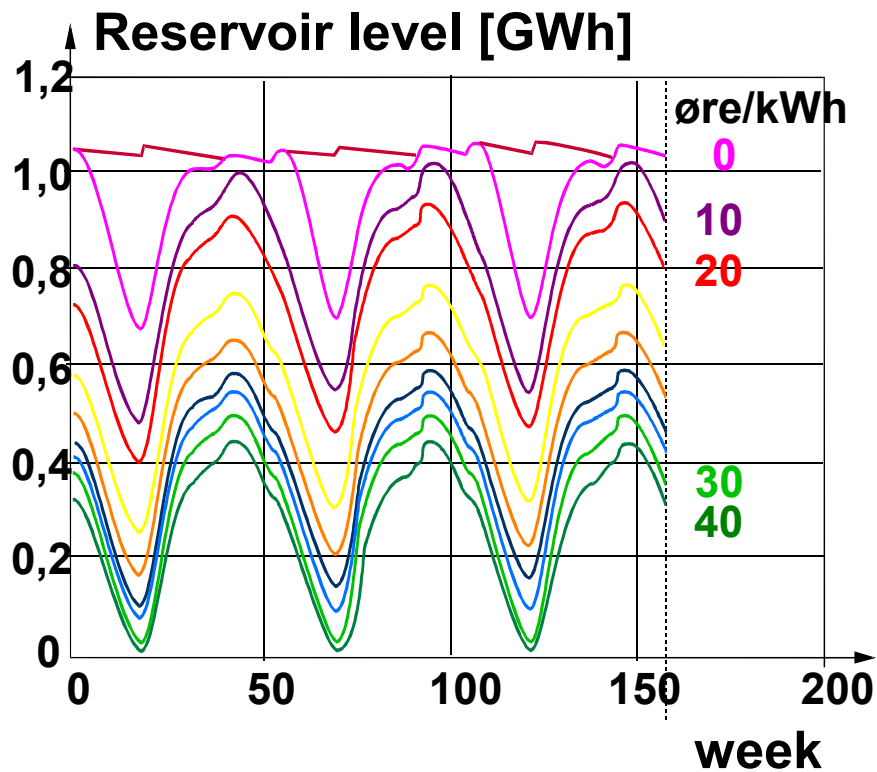


Water value calculations for discrete reservoir levels



Water value example

iso-price curves



Conclusions

- ❖ Hydropower scheduling operates within a hierarchical framework:
 - ✓ Long-term → strategic resource planning
 - ✓ Short-term → operational optimization and market participation
- ❖ The water value represents the marginal value of stored energy, the cornerstone linking economic and physical decision layers.
- ❖ Stochastic Dynamic Programming (SDP) provides a systematic way to derive water values through backward recursion over discretized reservoir states and inflow scenarios.
- ❖ Final water values provide a foundation for bidding strategies, system planning, and portfolio optimization in hydro-dominated markets